Design and Simulation of a Current-Mirror Bandgap Voltage Reference in SCL 180 nm CMOS

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Abstract

This report documents the design and schematic-level simulation of a self-biased, current-mirror based bandgap reference (BGR) in the SCL 180 nm process (`ts018_scl_prim`) using Cadence Virtuoso and Spectre. The circuit was built up in stages: a single vertical PNP to obtain the CTAT voltage, a pair of PNPs with ideal current sources to obtain the PTAT voltage $V_T \ln n$, a self-biased PTAT core with a PMOS/NMOS current mirror, and finally the complete reference where a mirrored PTAT current flows through a resistor R_2 in series with a third PNP. R_2 was then tuned with a parametric sweep. With $R_0 = 2.8\,\text{k}\Omega$ and $R_2 \approx 68.2\,\text{k}\Omega$ the reference gives $V_{ref} \approx 1.14\,\text{V}$ with a peak-to-peak variation of 13.34 mV over -40 to $125\,^\circ\text{C}$, i.e. a temperature coefficient of about 71 ppm/°C, from a 3.3 V supply.

1 Introduction

Almost every analog and mixed-signal block (ADCs, DACs, LDOs, PLLs, bias generators) needs a reference voltage that does not move with temperature or supply. A diode or a V_{GS} cannot do this on its own because both are strongly temperature dependent. The bandgap reference solves the problem by adding two voltages with opposite temperature coefficients: the base-emitter voltage of a bipolar transistor, which falls with temperature (CTAT), and a scaled difference of two base-emitter voltages, which rises with temperature (PTAT). If the scaling is chosen correctly the two slopes cancel and the output settles close to the silicon bandgap voltage, around 1.2 V.

The work follows the current-mirror BGR discussed in the lecture series by Hafeez K. T. [1], which was originally simulated in a 0.35 µm process. Here the same flow is repeated in SCL 180 nm, so the device choices, the resistor values and some of the results differ from the lecture, and those differences are discussed in Section 5.

2 Theory

2.1 CTAT voltage: base-emitter voltage of a BJT

For a forward-active BJT,

$$I_C = I_S \exp\!\left(\frac{V_{BE}}{V_T}\right) \quad \Rightarrow \quad V_{BE} = V_T \ln \frac{I_C}{I_S}, \qquad V_T = \frac{kT}{q}. \tag{1}$$

The saturation current depends strongly on temperature,

$$I_S = b\, T^{4+m} \exp\!\left(\frac{-E_g}{kT}\right), \tag{2}$$

where b is a constant, $m \approx -3/2$ comes from the temperature dependence of mobility, and $E_g \approx 1.12\,\text{eV}$ is the bandgap energy of silicon. Differentiating (1) with I_C held constant:

$$\frac{\partial V_{BE}}{\partial T} = \frac{V_T}{T}\ln\frac{I_C}{I_S} - \frac{V_T}{I_S}\frac{\partial I_S}{\partial T}. \tag{3}$$

From (2),

$$\frac{\partial I_S}{\partial T} = b(4+m)T^{3+m}e^{-E_g/kT} + b\,T^{4+m}e^{-E_g/kT}\frac{E_g}{kT^2} \;\Rightarrow\; \frac{V_T}{I_S}\frac{\partial I_S}{\partial T} = (4+m)\frac{V_T}{T} + \frac{E_g}{qT}. \tag{4}$$

Substituting back and using $V_T \ln(I_C/I_S) = V_{BE}$:

$$\boxed{\frac{\partial V_{BE}}{\partial T} = \frac{V_{BE} - (4+m)V_T - E_g/q}{T}} \tag{5}$$

With $V_{BE} \approx 675\,\text{mV}$ at $300\,\text{K}$ (the simulated value in Section 4.1), $m = -1.5$ and $V_T = 25.9\,\text{mV}$,

$$\frac{\partial V_{BE}}{\partial T} \approx \frac{0.675 - 2.5(0.0259) - 1.12}{300} \approx -1.70\,\text{mV/K}.$$

If the BJT is biased with a PTAT current instead of a constant one ($I_C \propto T$), the term $V_T \ln I_C$ contributes an extra $+V_T/T$, so $(4+m)$ in (5) becomes $(3+m)$ and the slope becomes slightly less negative (about $-1.61\,\text{mV/K}$ here). This is the case for Q_3 in the final circuit.

2.2 PTAT voltage: difference of two base-emitter voltages

If two BJTs carry the same current I but the second one has n times the emitter area (implemented as n unit devices in parallel), its saturation current is nI_S. Then

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln\frac{I}{I_S} - V_T \ln\frac{I}{nI_S} = V_T \ln n, \tag{6}$$

$$\frac{\partial \Delta V_{BE}}{\partial T} = \frac{k}{q}\ln n. \tag{7}$$

The result is independent of I and of I_S, which is what makes it a clean PTAT quantity. For $n = 2$, $\Delta V_{BE} = 17.9\,\text{mV}$ at $300\,\text{K}$ and the slope is $86.2\,\mu\text{V/K} \times \ln 2 = 59.7\,\mu\text{V/K}$.

2.3 Self-biased PTAT current generator

In the current-mirror core (Fig. 6) the NMOS pair M_1, M_2 shares a common gate, and the PMOS pair M_0, M_5 forces equal currents in the two branches. If M_1 and M_2 are identical and carry equal currents, their V_{GS} are equal, so the sources sit at the same potential:

$$V_D = V_{D1} + IR_0. \tag{8}$$

The left branch has one PNP ($V_D = V_{BE}$ at current I) and the right branch has n PNPs in parallel (V_{D1}). Using (6),

$$IR_0 = V_D - V_{D1} = V_T \ln n \quad\Rightarrow\quad \boxed{I = \frac{V_T \ln n}{R_0}} \tag{9}$$

The branch current is therefore PTAT and is set only by R_0 and n; transistor sizes and V_{DD} do not appear to first order. With $n = 2$ and $R_0 = 3.6\,\text{k}\Omega$, $I = 17.9\,\text{mV}/3.6\,\text{k}\Omega \approx 5\,\mu\text{A}$ at room temperature.

This loop also has a second stable operating point at $I = 0$ (all devices off). A practical design needs a start-up circuit to leave this state; that is listed as future work.

2.4 Complete bandgap reference

A third PMOS M_4 with the same gate voltage copies I into a resistor R_2 in series with a PNP Q_3. The output is

$$V_{ref} = V_{BE3} + IR_2 = V_{BE3} + \frac{R_2}{R_0} V_T \ln n. \tag{10}$$

Setting $\partial V_{ref}/\partial T = 0$ gives the first-order condition

$$\frac{R_2}{R_0} = \frac{|\partial V_{BE3}/\partial T|}{(k/q)\ln n}. \tag{11}$$

With the PTAT-biased slope of about 1.61 mV/K and $n = 2$, $R_2/R_0 \approx 1.61/0.0597 \approx 27$. Because V_{BE} is not perfectly linear in T (there is a $T \ln T$ term hidden in (1) and (2)), $V_{ref}(T)$ is a parabola-like "bell" and only its slope at one temperature can be made zero. The final value of R_2 is therefore found by simulation.

The figure of merit used here is the box-method temperature coefficient,

$$TC = \frac{V_{ref,max} - V_{ref,min}}{V_{ref,avg}\,(T_{max} - T_{min})} \times 10^6 \ \text{ ppm/°C}. \tag{12}$$

3 Simulation setup

Table 1: Process, devices and simulation settings.

Item	Value
Process / library	SCL 180 nm, `ts018_scl_prim`
Supply	$V_{DD} = 3.3$ V (`vdc`)
MOSFETs	`pmos_33`, `nmos_33` (3.3 V thick-oxide), $W/L = 20\,\mu\text{m}/5\,\mu\text{m}$
Bulk connections	PMOS bulk to V_{DD}, NMOS bulk to ground
BJTs	`vpnp33_10` (vertical PNP, $10 \times 10\,\mu$m emitter), base and collector grounded
Area ratio	$n = 2$ ($Q_1 \parallel Q_2$)
Simulator	Spectre (ADE L), DC sweep of temperature -40 to 125 °C

The 3.3 V devices were used throughout because the 1.8 V core devices of this process are not rated for a 3.3 V supply. Long channels ($L = 5\,\mu$m) were chosen to reduce channel-length modulation in the mirrors.

Note on schematic annotations: the node voltages printed on the schematic screenshots are the DC values at the first point of the temperature sweep, i.e. at -40 °C.

4 Simulations and results

4.1 Sim 1: CTAT voltage of a single PNP

A 5 µA ideal current source biases one `vpnp33_10` (Fig. 1) and the emitter voltage is swept over temperature.

The voltage falls almost linearly from 790 mV to about 507 mV, a slope of $(507 - 790)/165 \approx -1.72$ mV/°C. The interpolated value at 27 °C is about 675 mV. Equation (5) with this V_{BE} predicts -1.70 mV/K, which agrees within about 1%.

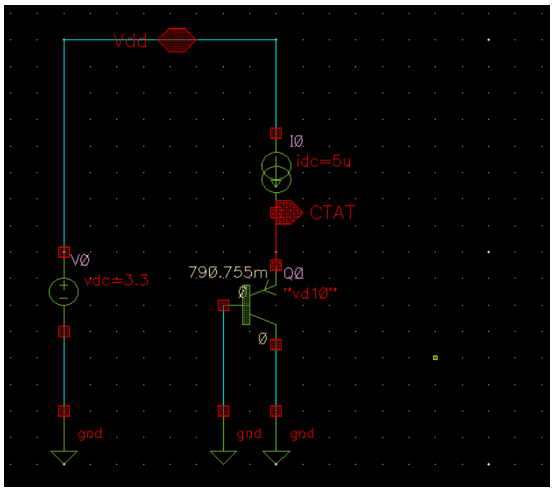


Figure 1: CTAT test circuit: $I_0 = 5\,\mu\text{A}$ into a diode-connected vertical PNP.

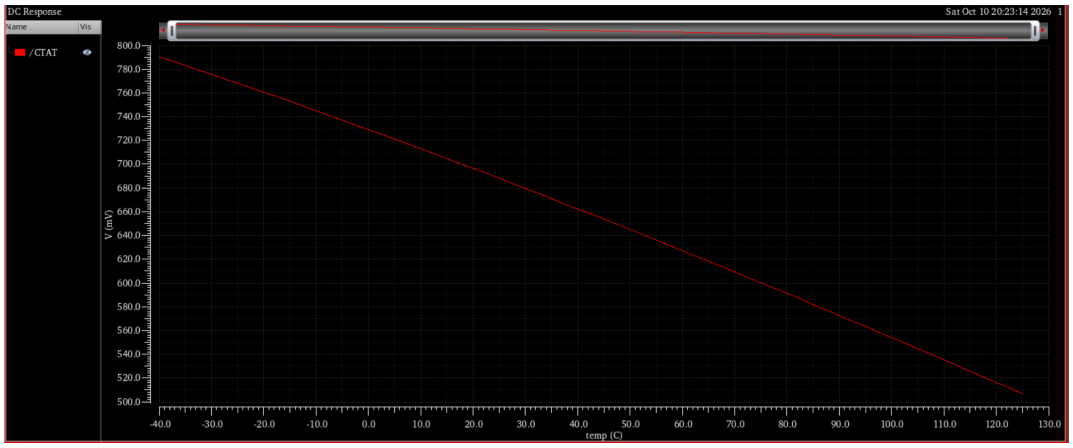


Figure 2: Emitter voltage versus temperature: 790 mV at $-40\,^\circ$C falling to about 507 mV at 125 °C.

4.2 Sim 2: PTAT voltage with ideal current sources

Two branches are biased with equal 5 µA ideal sources: one with a single PNP (V_D) and one with two PNPs in parallel (V_{D1}), Fig. 3.

The difference rises linearly from 13.5 mV to 23.3 mV, a slope of 59.4 µV/°C, against the theoretical $(k/q)\ln 2 = 59.7\,\mu\text{V/K}$. A marker at 2.9 °C reads 16.18 mV; the ideal value $V_T \ln 2$ at 276 K is 16.5 mV.

4.3 Sim 3: Self-biased PTAT core with current mirror

The ideal sources are replaced by the PMOS mirror (M_0, M_5) and the NMOS pair (M_1, M_2), with $R_0 = 3.6\,\text{k}\Omega$ in the $n = 2$ branch (Fig. 6). M_5 and M_2 are diode connected. A third PMOS M_4 copies the current into $R_1 = 100\,\text{k}\Omega$ to ground so that the PTAT current can be observed as a voltage.

At $-40\,^\circ$C the annotated values give $V_D = 785.8$ mV, source of $M_1 = 786.3$ mV and $V_{D1} = 772.3$ mV. The drop across R_0 is 14.0 mV, essentially equal to $V_T \ln 2 = 13.9$ mV at 233 K, which confirms that the loop settles where (9) predicts. The current is $14.0/3.6 \approx 3.9\,\mu\text{A}$ at $-40\,^\circ$C and about 5.1 µA at 27 °C (510 mV/100 kΩ), against 4.98 µA from (9).

The measured slope across R_1 is 1.79 mV/°C, about 8% above the ideal $(R_1/R_0)(k/q)\ln 2 =$

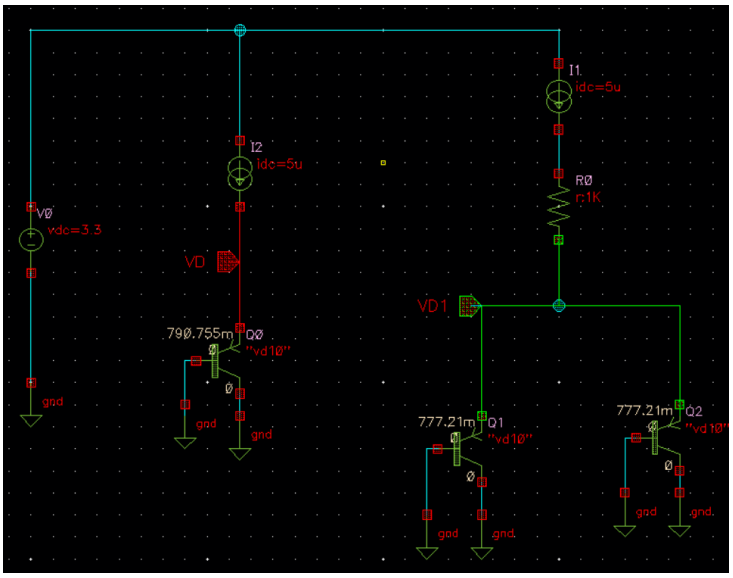


Figure 3: PTAT test circuit with ideal current sources ($n = 2$).



Figure 4: V_D (one PNP) and V_{D1} (two PNPs) versus temperature. Both are CTAT, but the gap between them grows with temperature.

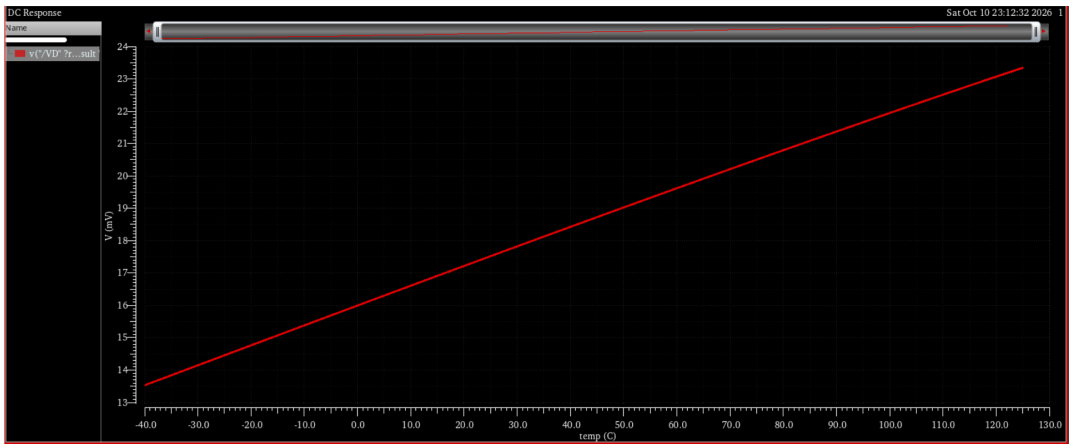


Figure 5: $V_D - V_{D1}$ versus temperature: 13.5 mV at $-40\,^\circ$C to 23.3 mV at $125\,^\circ$C.

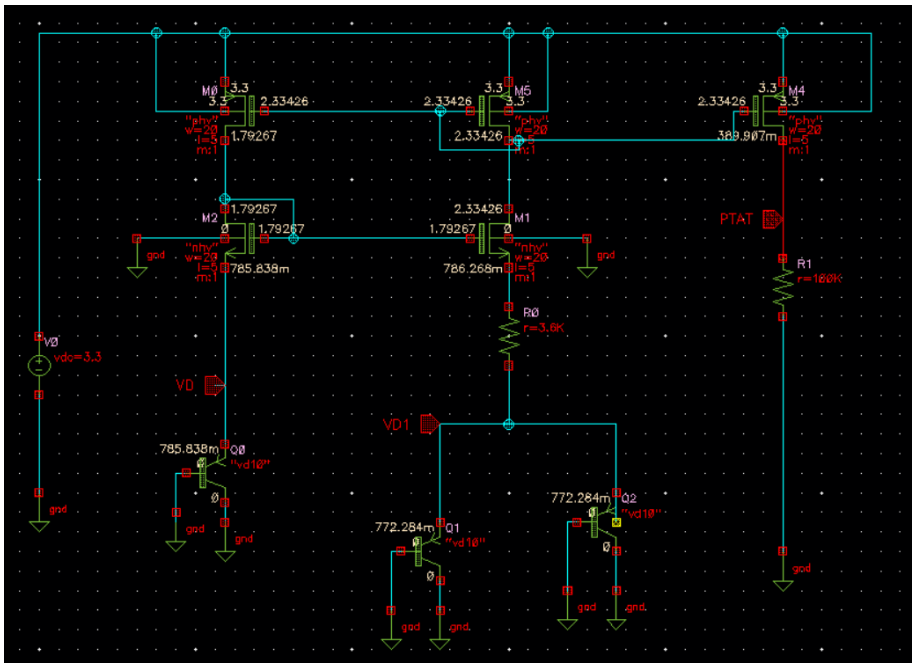


Figure 6: Self-biased PTAT core with an output copy through M_4 into $R_1 = 100\,\mathrm{k\Omega}$.

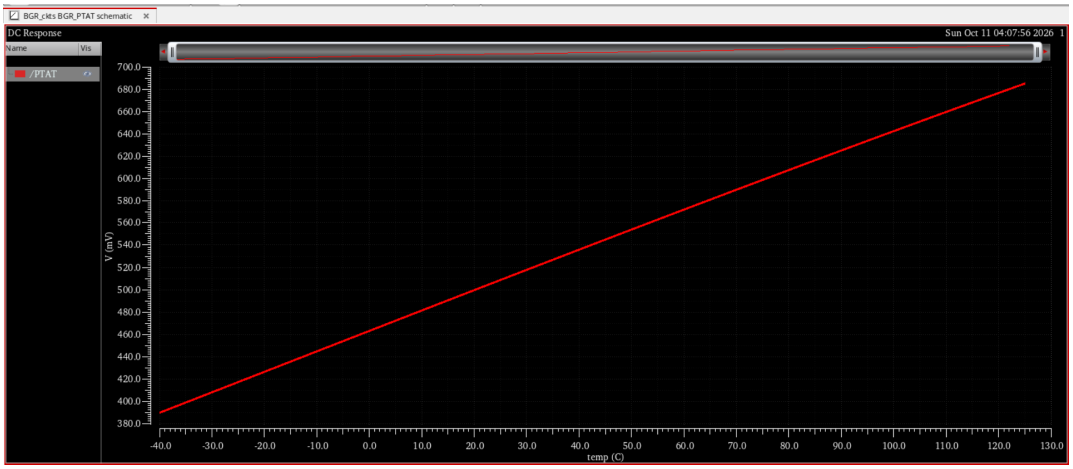


Figure 7: Voltage across R_1: about 390 mV at $-40\,^\circ$C rising to 685 mV at $125\,^\circ$C.

1.66 mV/K. The likely reason is channel-length modulation in M_4: its drain sits much lower than the drain of the diode-connected M_5, so it copies slightly more current than the reference branch.

4.4 Sim 4: Complete bandgap reference and tuning of R_2

R_1 is now connected to the emitter of a third PNP Q_3 instead of ground, giving the full reference of Fig. 8. For the final design R_0 was reduced to 2.8 kΩ, which sets $I \approx 5\,\mu\mathrm{A}$ at $-40\,^\circ$C and about 6.4 µA at $27\,^\circ$C. The output resistor (labelled R_1 on the schematic, called R_2 in the equations) was made a design variable.

The first run already shows the expected bell shape, but the peak sits near $57\,^\circ$C, so the PTAT term is slightly too strong. R_2 was therefore swept with a parametric analysis from 64 kΩ to 84 kΩ in 20 steps (Fig. 10). Larger R_2 moves the curve up and makes it rise with temperature; smaller R_2 makes it droop at high temperature. The flattest, most symmetric curve (highlighted) corresponds to $R_2 \approx 68.2\,\mathrm{k\Omega}$.

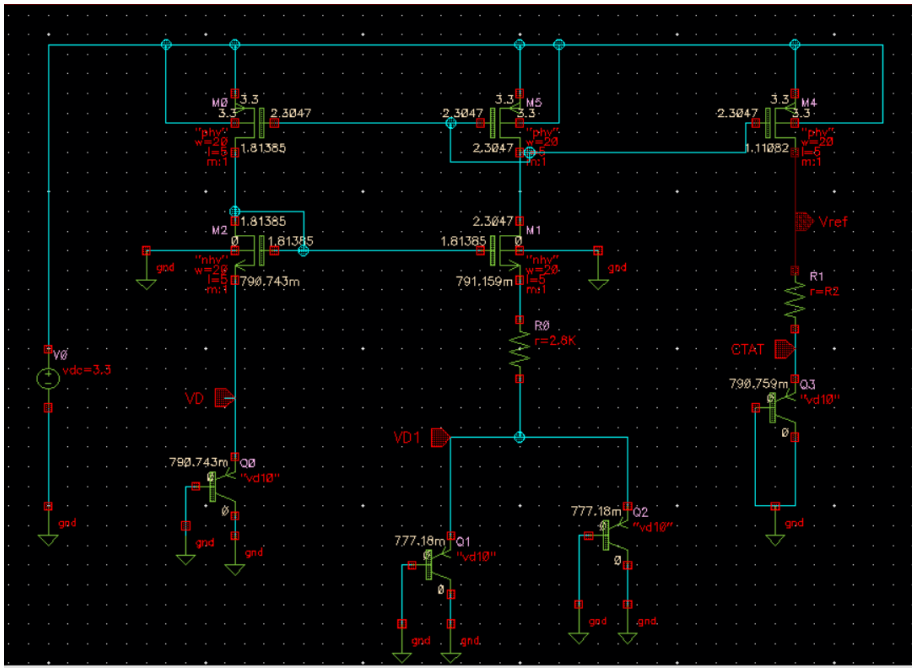


Figure 8: Complete current-mirror bandgap reference. $R_0 = 2.8\,\mathrm{k\Omega}$; the output resistor is the design variable R_2.

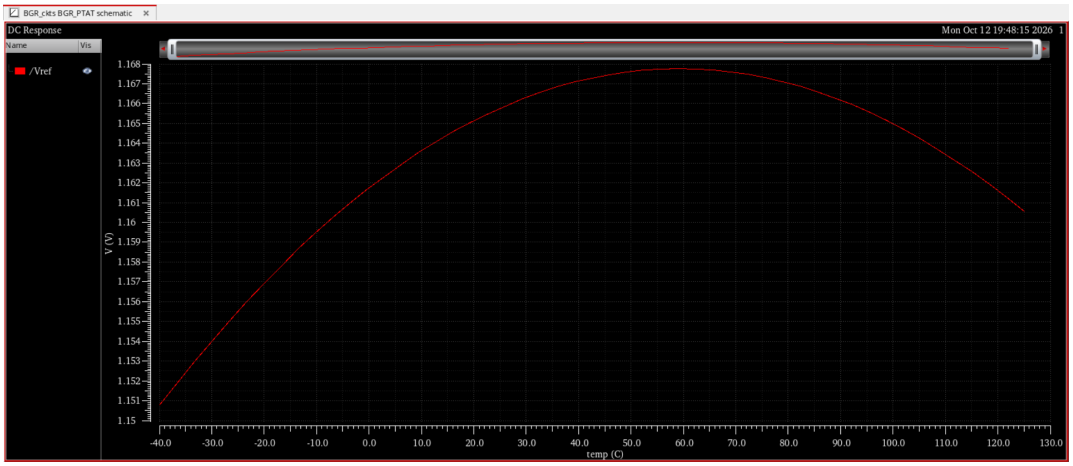


Figure 9: First run with an untuned R_2: a bell curve already appears, with the peak shifted to about $57\,^{\circ}\mathrm{C}$ and a spread of about $17\,\mathrm{mV}$.

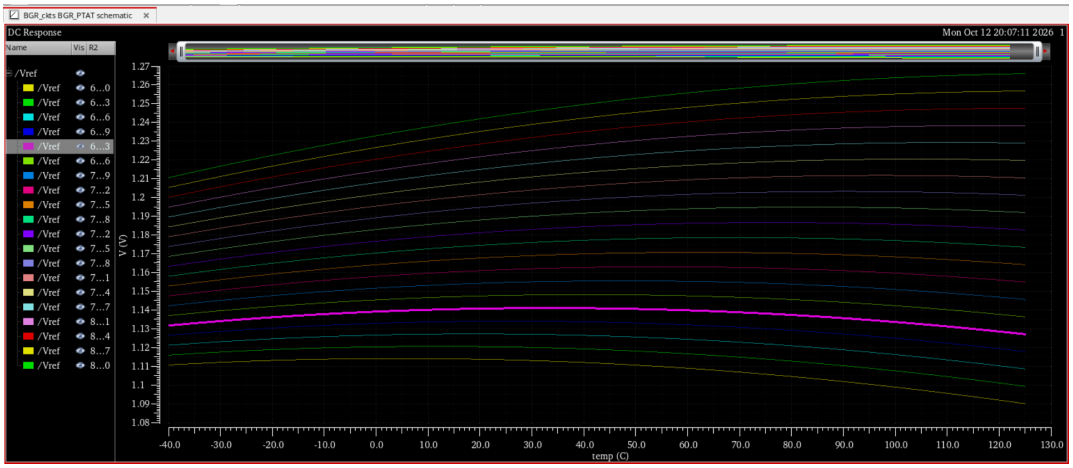


Figure 10: Parametric sweep of R_2 (64 to $84\,\mathrm{k\Omega}$). The highlighted trace is the selected value.

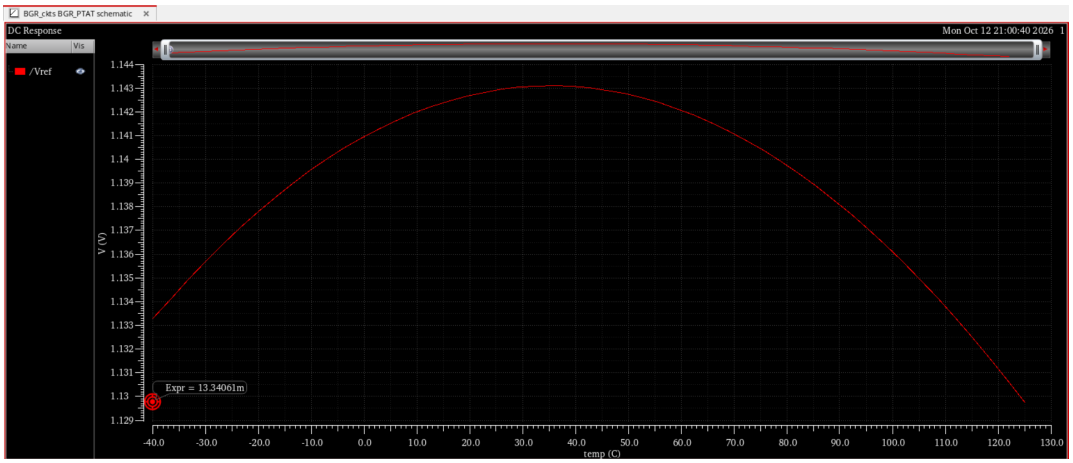


Figure 11: Final V_{ref} with $R_2 \approx 68.2\,\text{k}\Omega$. The calculator expression `ymax(VS("/Vref")) - ymin(VS("/Vref"))` returns 13.34 mV.

For the tuned circuit V_{ref} is 1.1333 V at $-40\,^\circ$C, peaks at 1.1431 V near $36\,^\circ$C and falls to 1.1298 V at $125\,^\circ$C. With $\Delta V = 13.34\,\text{mV}$ and $V_{ref,avg} \approx 1.139\,\text{V}$, equation (12) gives

$$TC = \frac{13.34 \times 10^{-3}}{1.139 \times 165} \times 10^6 \approx 71\,\text{ppm}/^\circ\text{C}.$$

5 Discussion

Resistor ratio. The simulated optimum is $R_2/R_0 = 68.2/2.8 \approx 24.4$, about 10% below the first-order estimate of 27 from (11). The same 8–10% excess current seen in the M_4 branch in Sim 3 explains most of this: if M_4 delivers more current than the core, a smaller R_2 is enough to produce the required PTAT slope. The lecture saw the same kind of gap between the calculated and the tuned resistor, which is why R_2 is always finalised with a sweep.

Output level. $V_{ref} \approx 1.14\,\text{V}$ is a little below the 1.2 V usually quoted. The value depends on V_{BE3} at the operating current and on the exact cancellation point; the SCL vertical PNP has a different saturation current from the device used in the lecture, so a shift of a few tens of millivolts is expected.

Temperature coefficient. The remaining 13 mV spread is set by the curvature of V_{BE} and by the finite output resistance of the simple mirror, neither of which can be removed by changing R_2. A cascode mirror (equal drain voltages in all branches) and curvature compensation are the usual next steps.

6 Debugging notes

A few problems came up while building the circuit, and they were useful for understanding it.

1. **Undefined model (SFE-23).** The first run stopped with "instance Q0 is referencing an undefined model". The SCL model file had not been added in *Setup → Model Libraries*; adding it with the correct section fixed the error.
2. **Wrong device in the mirror.** In the first PTAT-core schematic the diode-connected reference transistor of the PMOS mirror had been placed as `nmos_33` instead of `pmos_33`

(visible in the Navigator list of Fig. 12). The output then *fell* with temperature and collapsed to zero near 100 °C (Fig. 13). With its body at V_{DD}, the NMOS was only conducting through its forward-biased body junction, so the "mirror reference" was really a diode whose drop decreases with temperature; the copied current fell until the loop dropped into its zero-current state. Replacing it with `pmos_33` restored a clean PTAT output.

3. **Units.** An ideal current source entered as `idc=5` is 5 A, not 5 µA; the emitter node then read 28 V. Values must be entered with the suffix (`5u`).

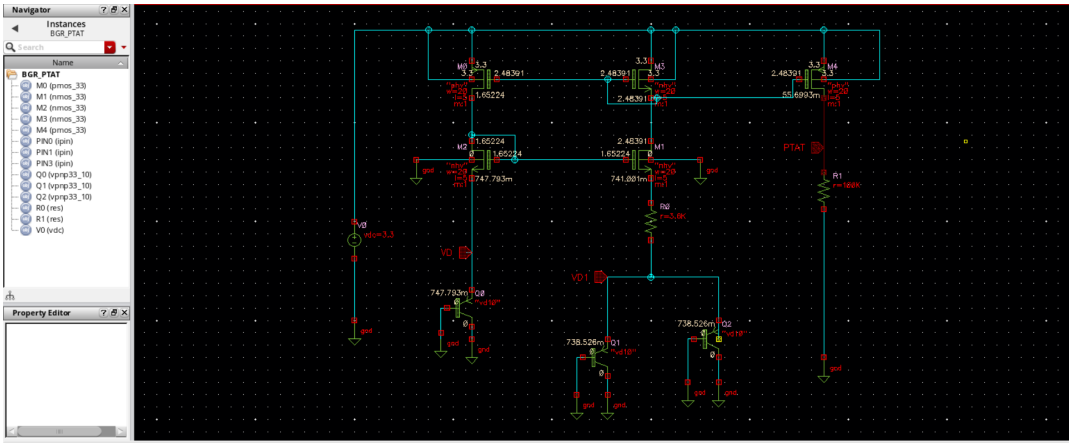


Figure 12: Faulty schematic: the Navigator shows M3 as `nmos_33` in the PMOS mirror position.

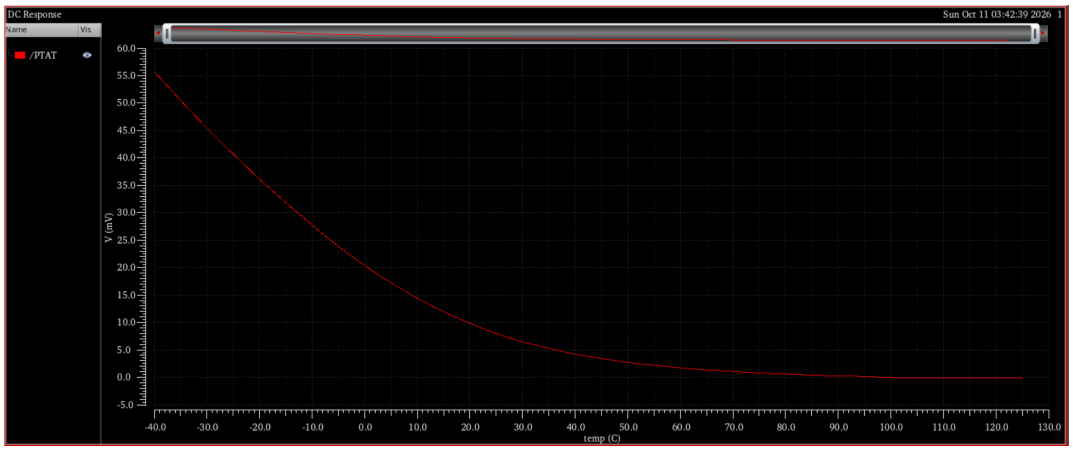


Figure 13: Output of the faulty circuit: the "PTAT" voltage decreases and reaches zero at high temperature.

7 Summary

Table 2: Summary of simulated results.

Quantity	Simulated	Expected / theory
CTAT slope ($I = 5\,\mu A$)	$-1.72\,mV/°C$	$-1.70\,mV/K$
V_{BE} at 27 °C	$\approx 675\,mV$	–
ΔV_{BE} slope ($n = 2$)	$59.4\,\mu V/°C$	$59.7\,\mu V/K$
ΔV_{BE} at 2.9 °C	$16.18\,mV$	$16.5\,mV$
PTAT core current, $R_0 = 3.6\,k\Omega$, 27 °C	$\approx 5.1\,\mu A$	$4.98\,\mu A$
Final R_0 / R_2	$2.8\,k\Omega$ / $\approx 68.2\,k\Omega$	ratio ≈ 27 (first order)
V_{ref} (peak, ≈36 °C)	$1.143\,V$	$\approx 1.2\,V$
ΔV_{ref} (−40 to 125 °C)	$13.34\,mV$	–
Temperature coefficient	$\approx 71\,ppm/°C$	–

8 Future work

- Supply sweep of V_{DD} (0 to 3.6 V) to find the minimum operating supply and the line regulation, first with the simple mirror and then with a cascode mirror.
- Start-up circuit and a transient power-up simulation to make sure the circuit never stays in the zero-current state.
- Op-amp based BGR topologies, process corners and Monte Carlo mismatch analysis.
- Layout with common-centroid PNP and mirror arrangements, followed by post-layout simulation.

References

[1] Hafeez K. T., lecture series on bandgap reference circuits (current-mirror and op-amp based BGR).

[2] B. Razavi, *Design of Analog CMOS Integrated Circuits*, 2nd ed., McGraw-Hill, 2017, Ch. 12.

[3] R. J. Widlar, "New developments in IC voltage regulators," *IEEE J. Solid-State Circuits*, vol. 6, no. 1, pp. 2–7, 1971.

[4] A. P. Brokaw, "A simple three-terminal IC bandgap reference," *IEEE J. Solid-State Circuits*, vol. 9, no. 6, pp. 388–393, 1974.